SageServe MLOps Platform

Forecast-aware autoscaling for LLM inference, and the MLOps layer that makes its forecasts trustworthy.

What is SageServe ?

A serving system

Autoscaler for large language model inference across several GPU regions and models.

Forecast-aware

It predicts incoming token demand and adds capacity before the load arrives.

Optimization-driven

A global MILP decides how many instances each region and model should run.

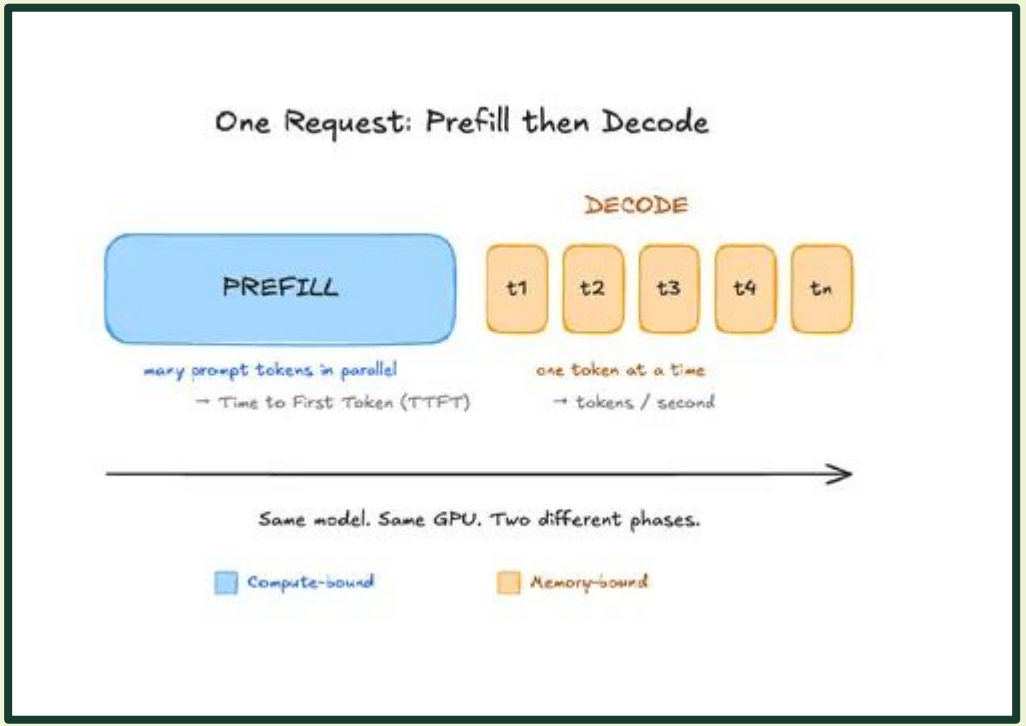
How LLM inference works

Prefill

Reads the whole prompt in parallel. Compute-bound. Fills the KV cache.

Decode

Generates one token per step. Memory-bandwidth-bound.



Why reactive autoscaling falls short ?

Bursty traffic

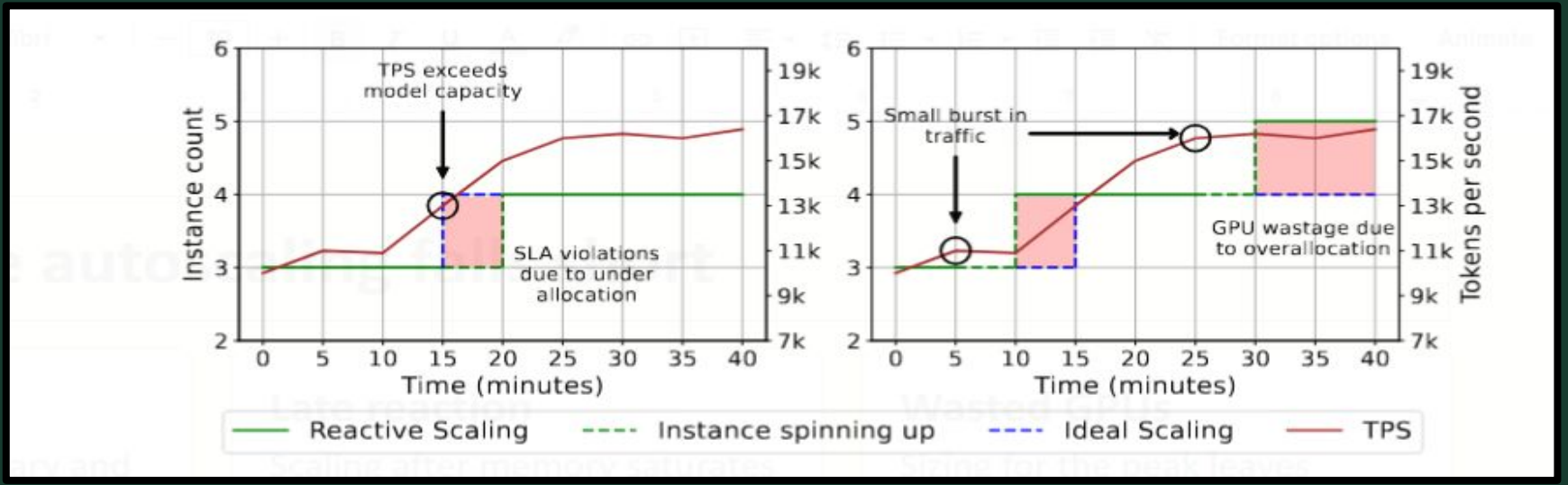
Demand is non-stationary and strongly diurnal.

Late reaction

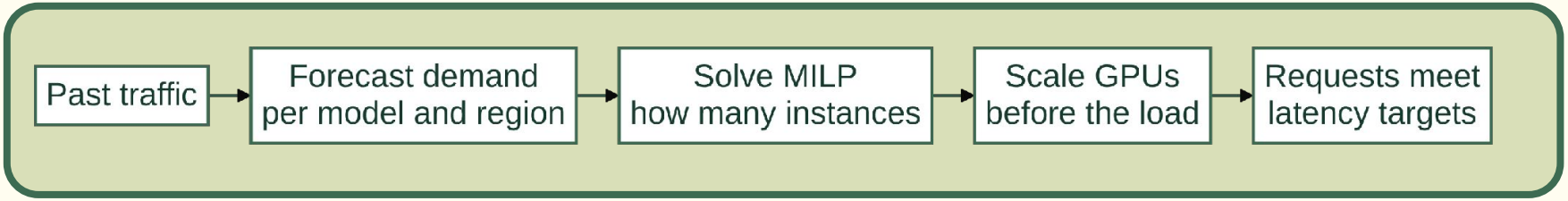
Scaling after memory saturates means queues, slow first tokens and SLA misses.

Wasted GPUs

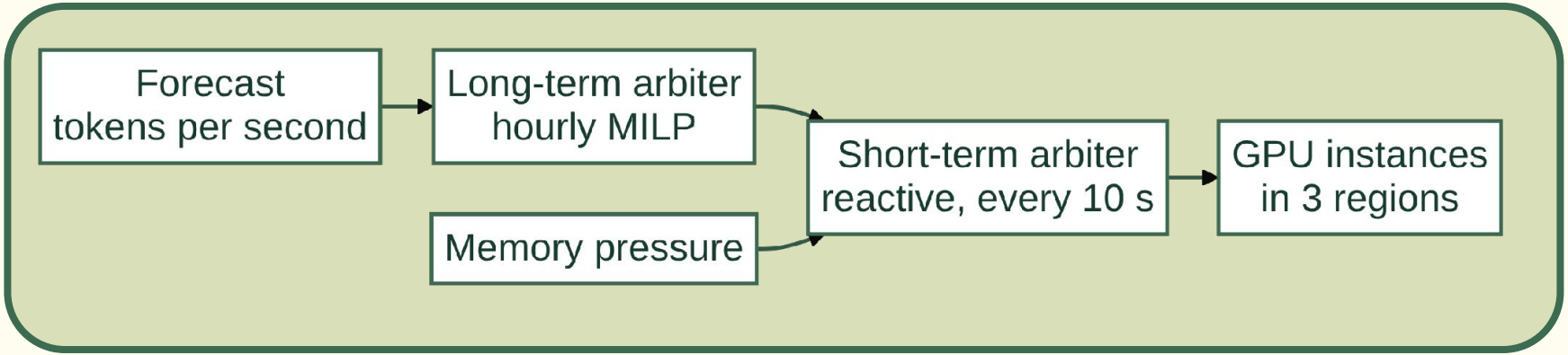
Sizing for the peak leaves expensive H100 / A100 GPUs idle.



The SageServe idea



Where the forecast fits



The gap: who looks after the forecaster

The forecast is an ML model

Research simulators often assume perfect or ad-hoc offline forecasts.

In production it needs care

Clean inputs, versions, approval, monitoring and repeatable runs.

QUESTIONS

- **Which data trained this forecast?**
- **Is the model approved for production?**
- **Can we reproduce this result?**
- **What can we infer ?**

What the MLOps platform adds

Data quality

Every trace is validated before use.

Versioning

Content hashes and a lineage ledger.

Model governance

Registry, model cards and promotion gates.

Monitoring

Latency, SLA and drift diagnostics.

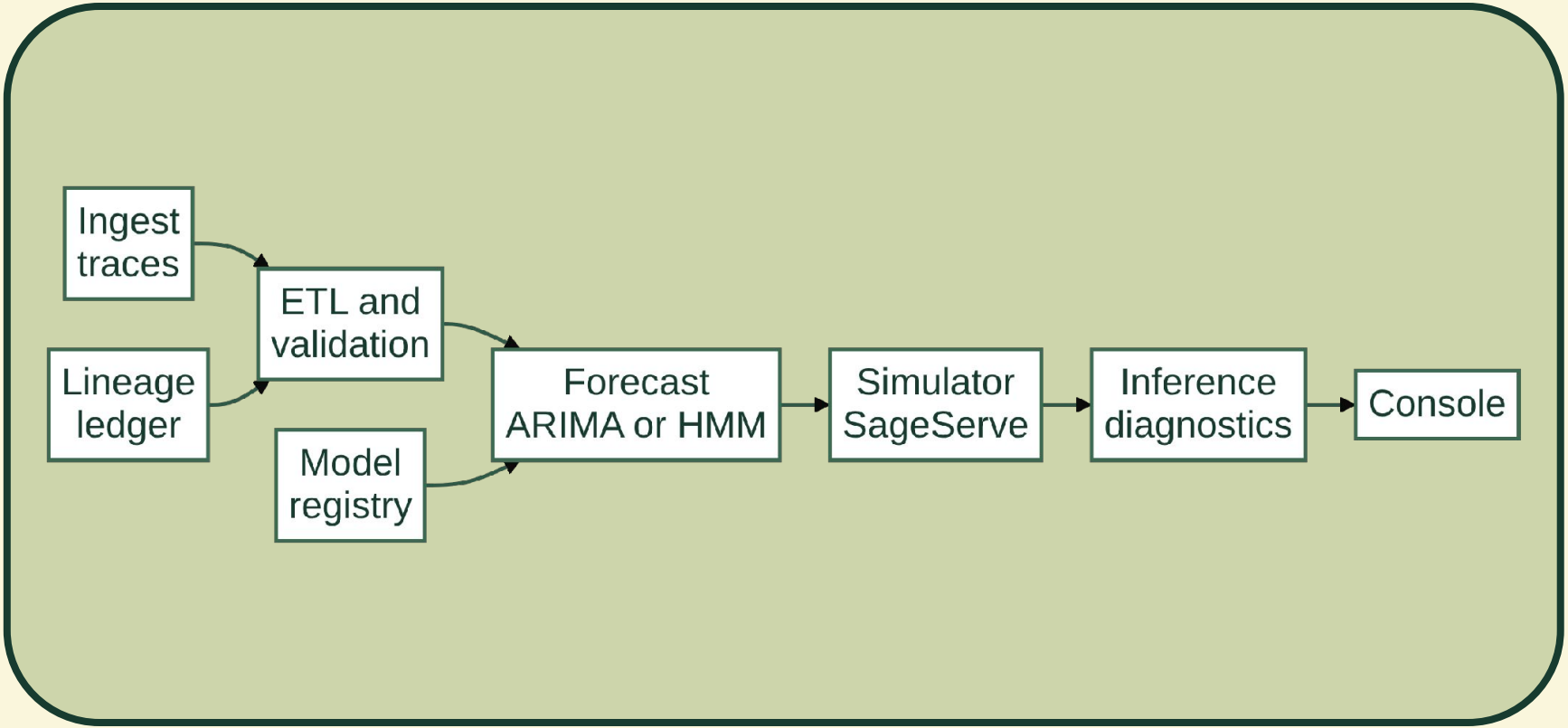
Reproducibility

Containers with isolated environments.

Automation

CI/CD runs the whole chain.

Platform architecture



Module 1: Trace Generator

Purpose

Produces realistic LLM traffic that follows the patterns described in the SageServe paper.

What it provides

Raw request traces in one strict 15-column format, identical byte for byte when the seed is the same.

What it does

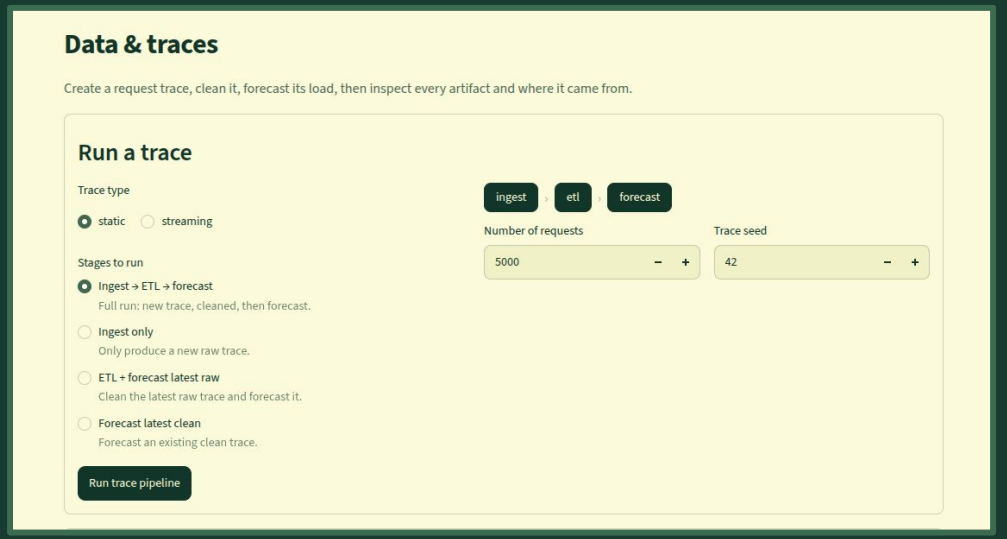
Generates diurnal request arrivals for 3 regions and 4 models across three SLA tiers, either as a seeded file or as a live HTTP stream that can be replayed faster than real time.

Why it matters

Every later stage starts from controlled, repeatable input. Since real production traces are not available.

The **Trace Generator** has two modes static and streaming

- Static is for random trace generation.
- Streaming for similar to paper trace generation



NOTE

The main objective of the project is the mlops pipeline not how the trace is generated.

Hence **Claude** was used to make the trace generator module.



Module 2: ETL Validation

Purpose

Makes sure no raw trace reaches a model without being checked.

What it provides

A validated trace with a manifest. If any check fails, the pipeline stops.

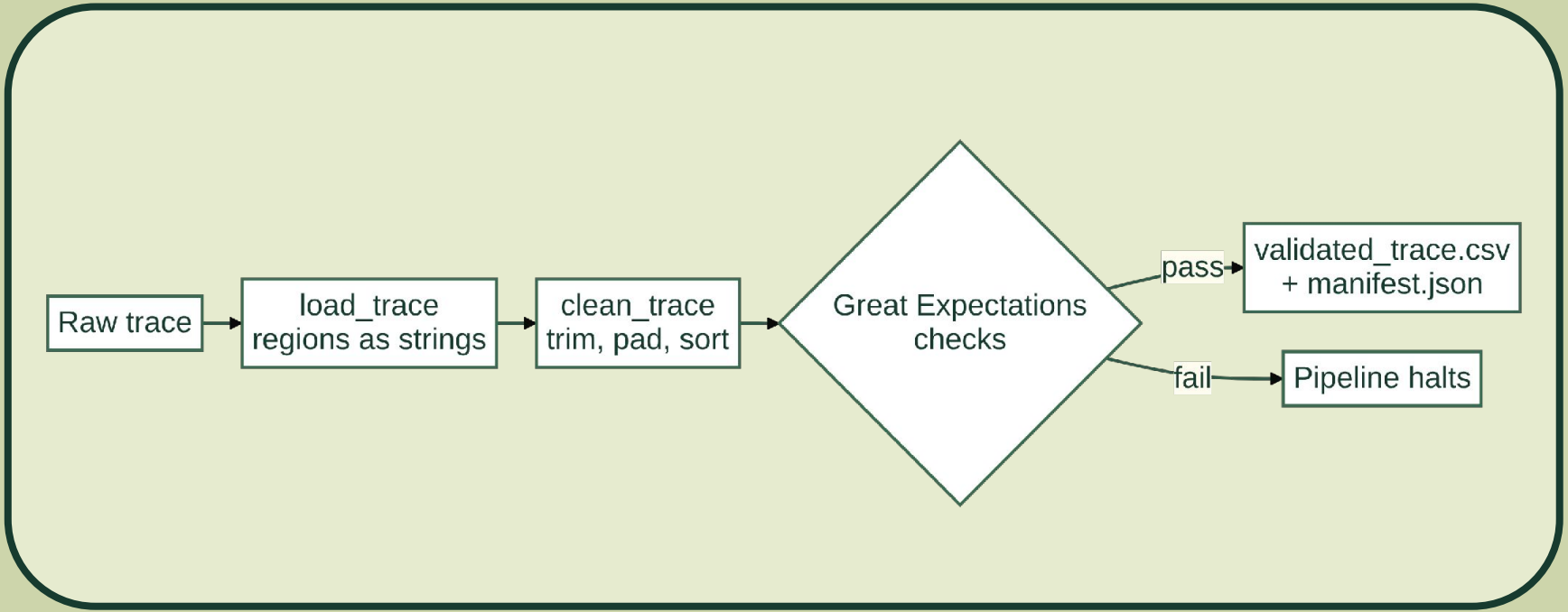
What it does

Standardizes types and formats, cleans the data, then validates it against a formal set of expectations covering schema, missing values, ranges and allowed categories.

Why it matters

Bad data is caught at the door, and a change in cleaning logic changes the version automatically.

ETL flow


Raw trace
load_trace
regions as strings
clean_trace
trim, pad, sort
Great Expectations
checks
pass
validated_trace.csv
+ manifest.json
fail
Pipeline halts

Module 3: Forecasting

Purpose

Predicts how many tokens each model will receive in each region, for the capacity planner.

What it provides

Forecast files per model and region, read by the simulator, behind one common interface for all models.

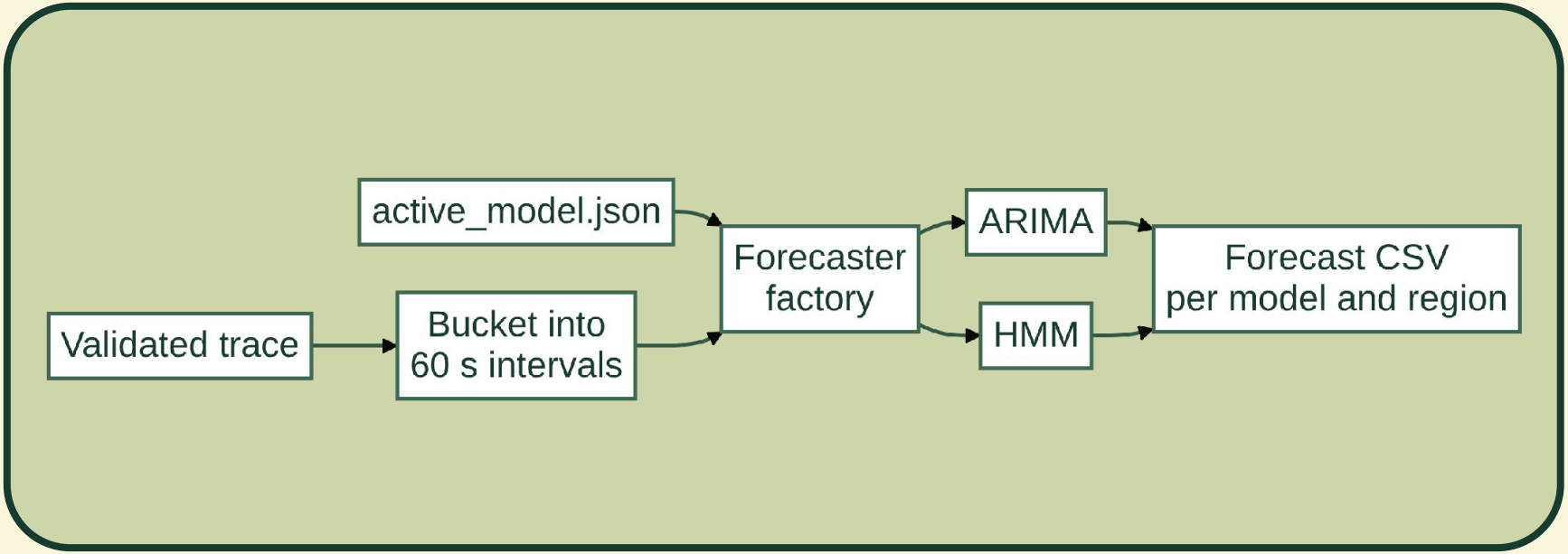
What it does

Turns the trace into per-minute demand, fits the active model (ARIMA or a Hidden Markov Model) and projects the next time horizon.

Why it matters

The registry decides which model runs, so models can be swapped without touching the pipeline.

Forecasting flow



Module 4: Model registry and governance

Purpose

Acts as the single source of truth and gatekeeper for forecasting models.

What it provides

One atomic pointer to the production model, and a command line to register, document and promote.

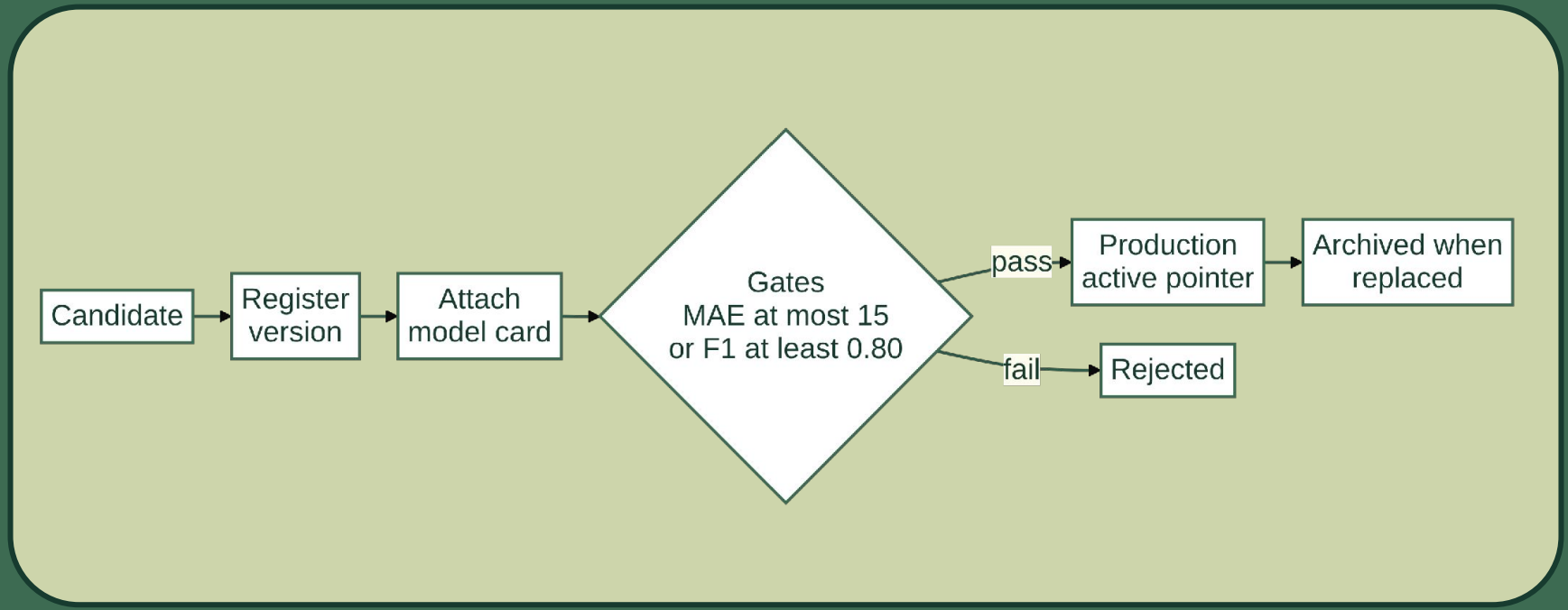
What it does

Stores every model version with its settings, metrics and a model card, and checks documentation and performance before a model can reach Production.

Why it matters

Only documented, qualified models serve traffic, and the history stays auditable.

Registry lifecycle



Forecast models

Register the models that predict load, document them, and decide which one is in Production.

Production model: **arima / v2**

Register a model

Model type

arima | hmm

Model description

ARIMA workload forecaster

Training window (minutes) 60 − +

Forecast horizon (minutes) 60 − +

Bucket size (seconds) 60 − +

AR order (p) 1 − +

Difference order (d) 1 − +

MA order (q) 1 − +

ARIMA is fitted to each workload series when you run a forecast; it has no saved weights file.

› Config file preview

Measured governance metrics

mae=12.4 or f1=0.84

Register ARIMA

Registered versions

Model	Version	Stage	Model card	Metrics
arima	v1	Archived	Ready	{"mae": 9.2}
arima	v2	Production	Ready	{"mae": 12.4}
arima	v3	None	Missing	{"mae": 8.5}
arima	v4	None	Missing	{"mae": 8.5}
arima	v5	None	Ready	{"mae": 8.5}
hmm	v1	None	Ready	{"mae": 11.0}
hmm	v2	Staging	Ready	{}
hmm	v3	Archived	Ready	{"f1": 0.89}

Manage version

arima / v1

› Registered config and manifest

Model card | Promote

Card description

Workload forecasting model

Intended use

Forecast per-minute SageServe request workload.

Limitations

Forecast quality depends on trace history and model parameters.

Save model card

Manage version

arima / v1

› Registered config and manifest

Model card | Promote

Promote to stage

Staging | Production | Archived

Promote model

Managing the Model Registry through web interface

Module 5: Pipeline, versioning and lineage

Purpose

Keeps runs deterministic, non-destructive and traceable.

What it provides

An append-only lineage ledger that traces any forecast back to its raw trace.

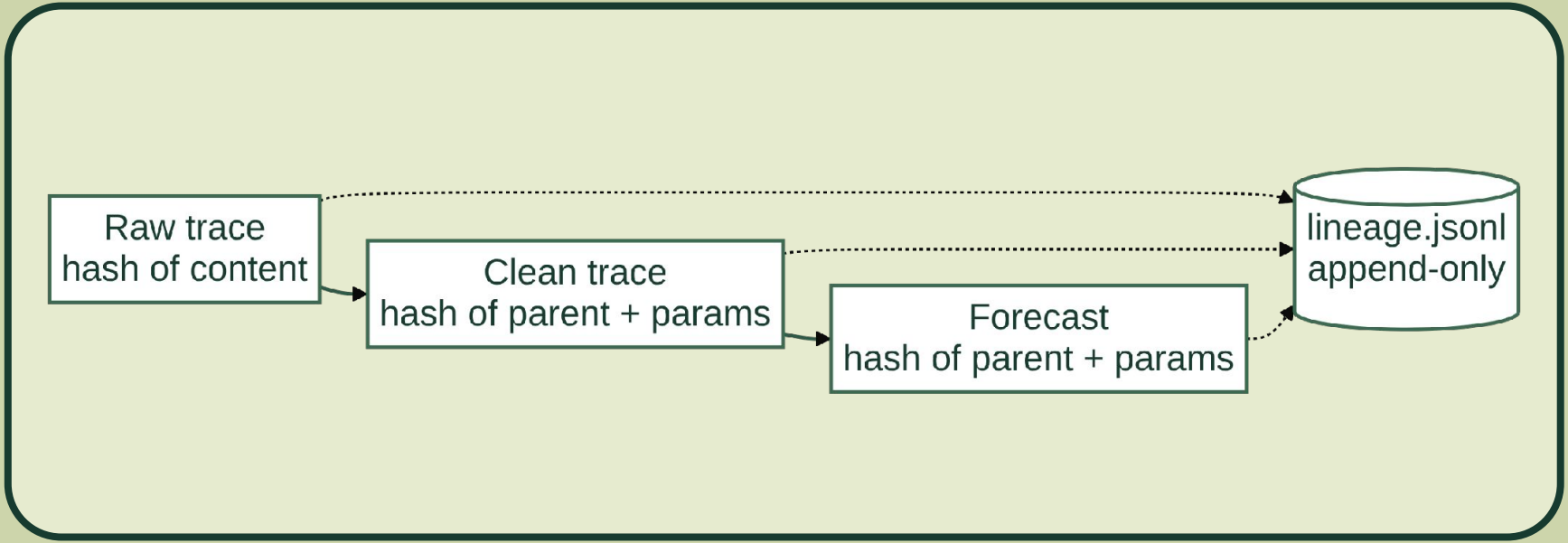
What it does

Runs ingest, ETL and forecast in order, gives each artifact a content-based version, skips work that already exists and records a manifest for each artifact.

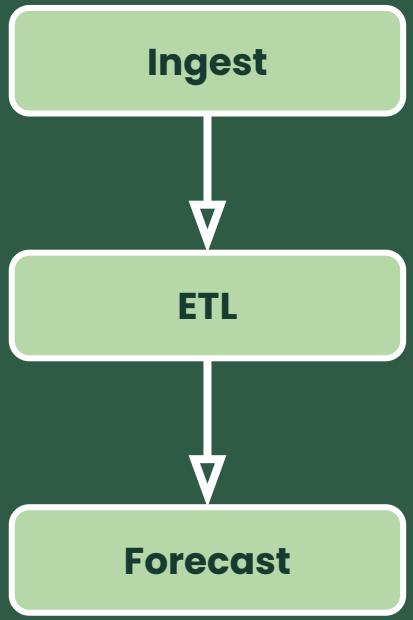
Why it matters

Results can be reproduced and audited without heavy external infrastructure.

Lineage flow



Pipeline Workflow



Browse data

Data source: static / streaming

Artifact stage: raw / clean / forecasts

Clean version: Latest · 425ea55de048

Version	Rows	Status	Created
425ea55d...	11522	available	2026-11-0...

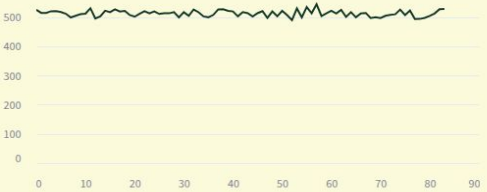
Artifact directory: `/workspace/data/streaming/clean/425ea55de048`

Data file: validated_trace.csv

Requests per minute



Mean prompt size per minute



Charts use up to the first 100,000 rows of the selected CSV.

Lineage

Artifact: streaming/forecasts/14573b794f71

Artifact	Created	Parent(s)	Stage
streaming/forecasts/14573b794f71	2026-11-09T06:25:43+00:00	streaming/clean/425ea55de048	forecasts
streaming/clean/425ea55de048	2026-11-09T06:25:42+00:00	streaming/raw/47f4f694fa4d	clean
streaming/raw/47f4f694fa4d	2026-11-09T06:25:37+00:00		raw

› Parameters and artifact details

Module 6: SageServe simulator

Purpose

Serves as the evaluation bed that shows what a forecast does to latency and cost.

What it provides

Logs of latency, instance counts and energy use for every run.

What it does

Replays traces through a multi-region GPU cluster, routes requests, models prefill and decode, and scales capacity in two layers: a reactive loop every 10 seconds and an hourly forecast-driven MILP.

Why it matters

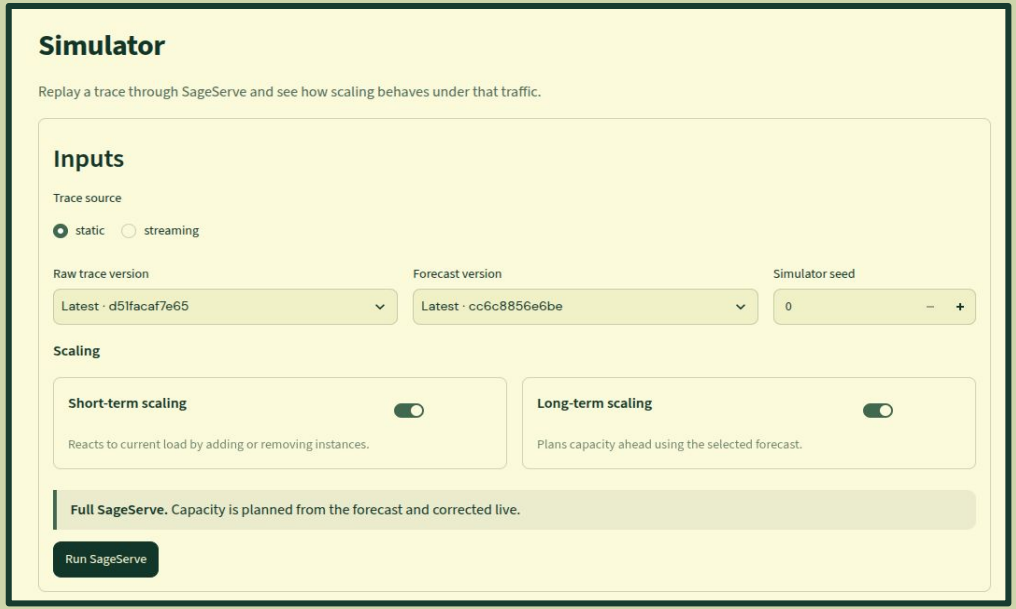
It measures the effect of each forecasting model on SLA violations and instance-hours.

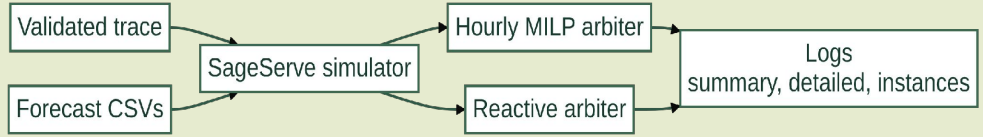
Simulator flow

We are using the simulator provided the original paper of the SageServe.

No changes is there implementation code has been done , only integration with there interface.

Refer :
https://github.com/shashwatj07/SageServe





Module 7: Inference and drift analysis

Purpose

Judges whether a simulation run was a success.

What it provides

Reports and latency CDF plots that compare configurations side by side.

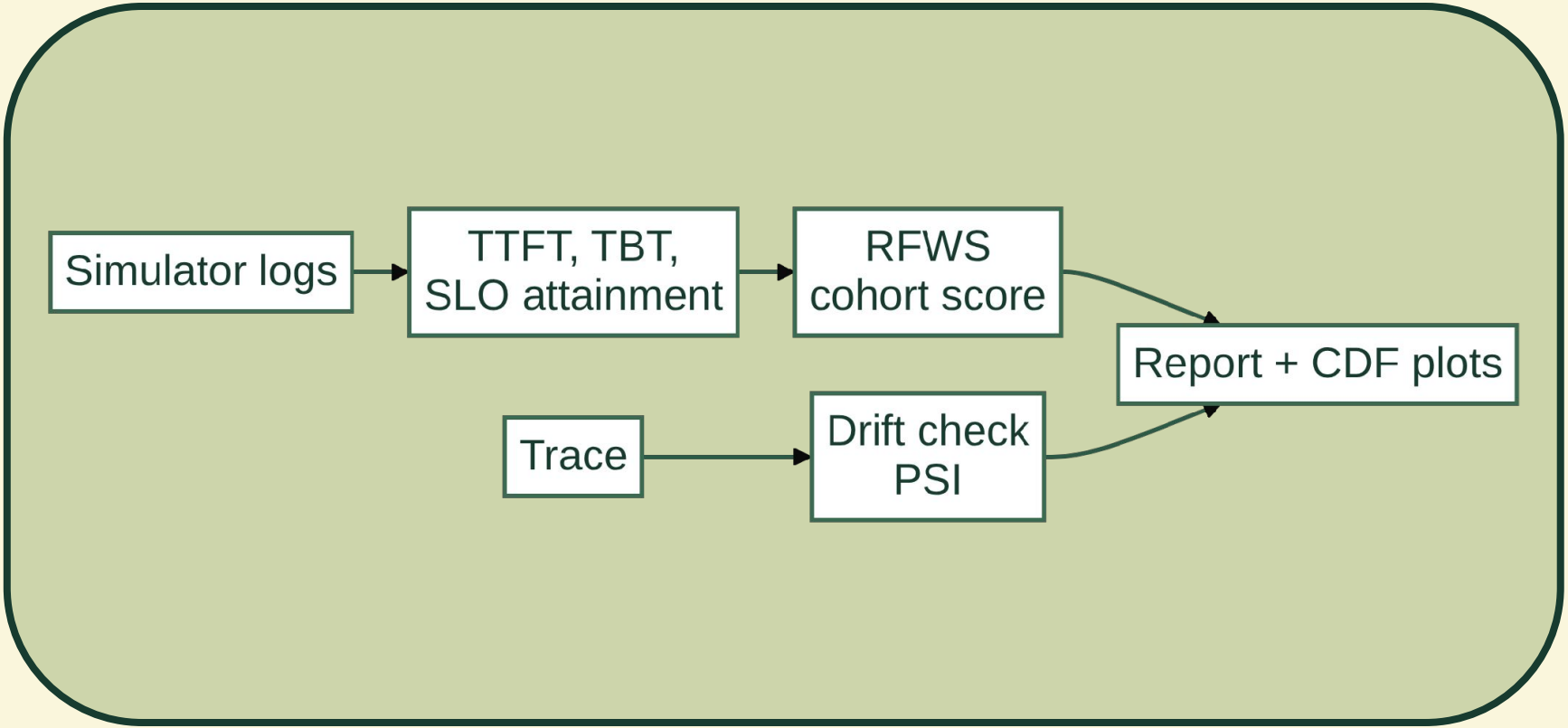
What it does

Loads the logs and computes TTFT and TBT violation rates, combines them into a cohort score, injects workload drift and evaluates forecasts on unseen data.

Why it matters

It turns raw telemetry into a decision about which model and settings to use.

Inference flow



Results

Analyze saved simulator output into latency and SLA reports. This does not run any LLM requests.

Generate a report

Simulator results directory

/workspace/experiments/results

Report name

sage_serve_run

Forecast model label

unverified

Allow partial input

Generate report

Reports

Report to view

sage_serve_run.csv

forecast_model	experiment_id	run	trace	lts	sts	arbiter	source_path
Arima	3f367290ed31f2ae	0	webui_static_20261108_181225	☑	☑	basic_arbiter	/workspace/experiments/results/0/webui_static_
Arima	3f367290ed31f2ae	0	webui_static_20261108_181225	☑	☑	basic_arbiter	/workspace/experiments/results/0/webui_static_
Arima	3f367290ed31f2ae	0	webui_static_20261108_181225	☑	☑	basic_arbiter	/workspace/experiments/results/0/webui_static_
Arima	3f367290ed31f2ae	0	webui_static_20261108_181225	☑	☑	basic_arbiter	/workspace/experiments/results/0/webui_static_

› Input diagnostics (4)

› Report provenance

Plot metric

e2e_p99 | viol_ttft | viol_avg_tbt



Completions by window

Experiment	Window	A completions	B completions	C completions	D completions
3f367290ed31f2ae	0	None	58	None	10669

One row per experiment and export window, with all models side by side. Window indexes are not timestamps. Blank counts are unavailable or incomplete; zero means a confirmed window with no completions.

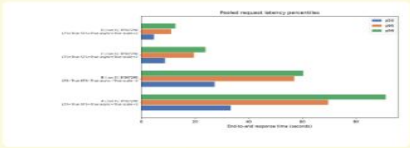
› Observed subsets from incomplete exports

Instance counts and allocation changes by region

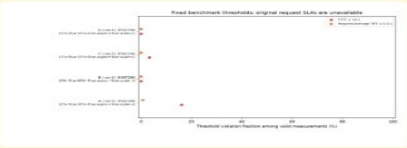
Allocation region

westus

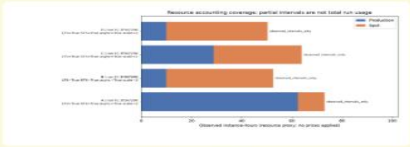




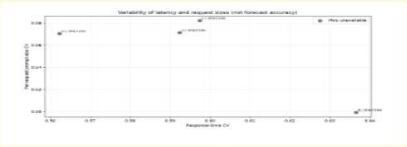
Latency Pct



Slo Violations



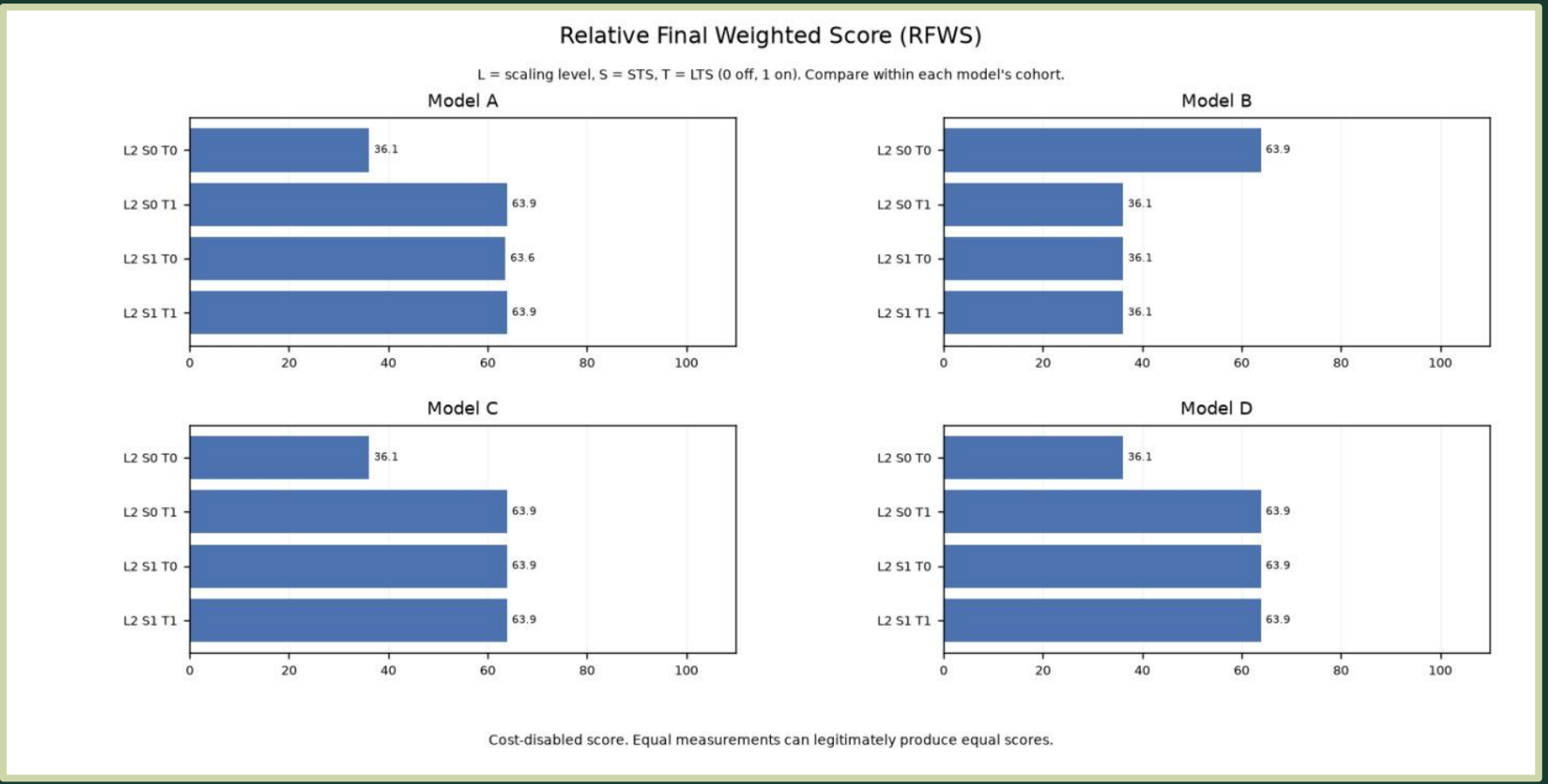
Resource Usage



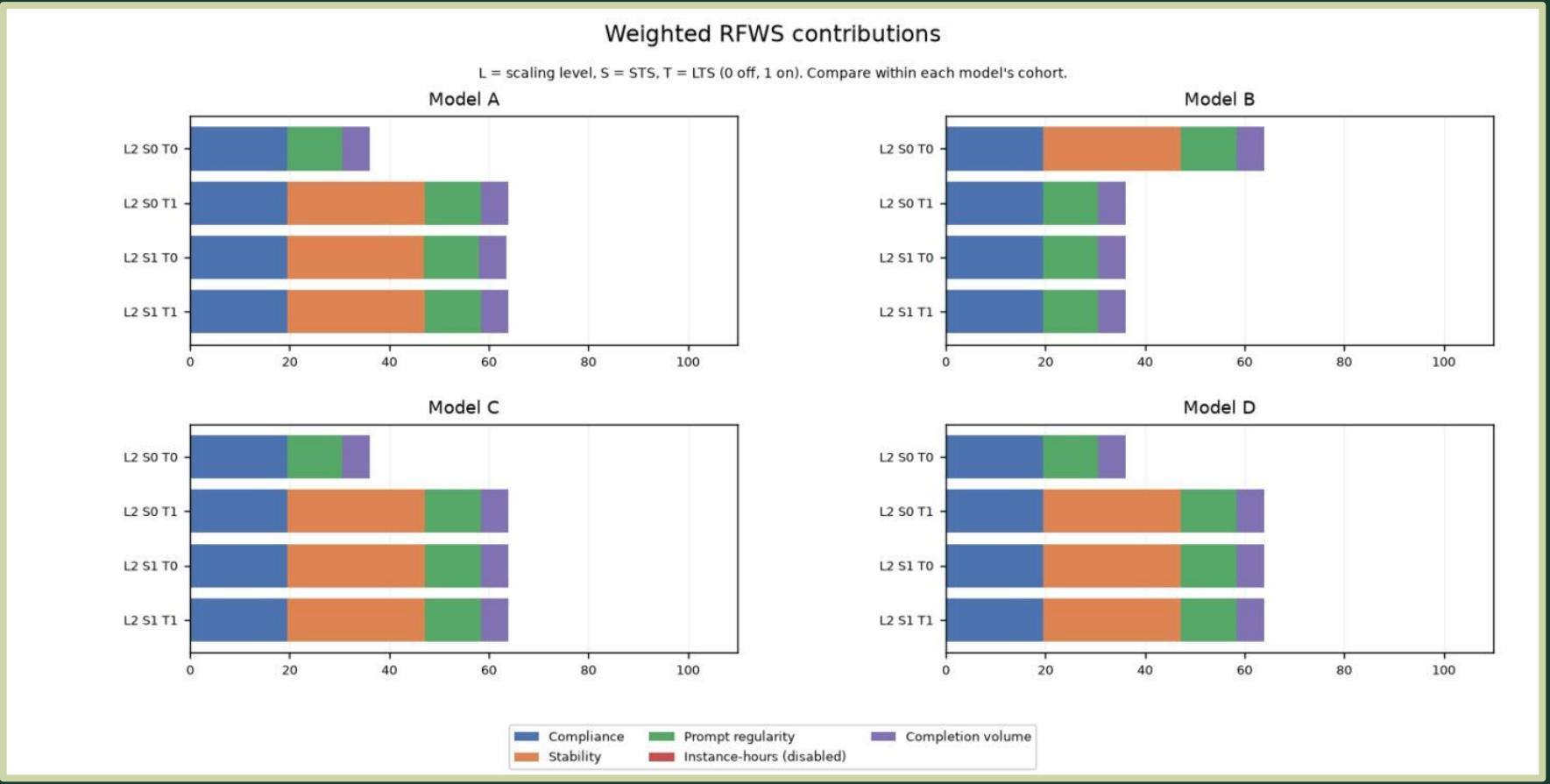
Cv Scatter

CSV: /workspace/mlops/inference/reports/sage_serve_run.csv · Plots: /workspace/mlops/inference/reports/sage_serve_run_plots

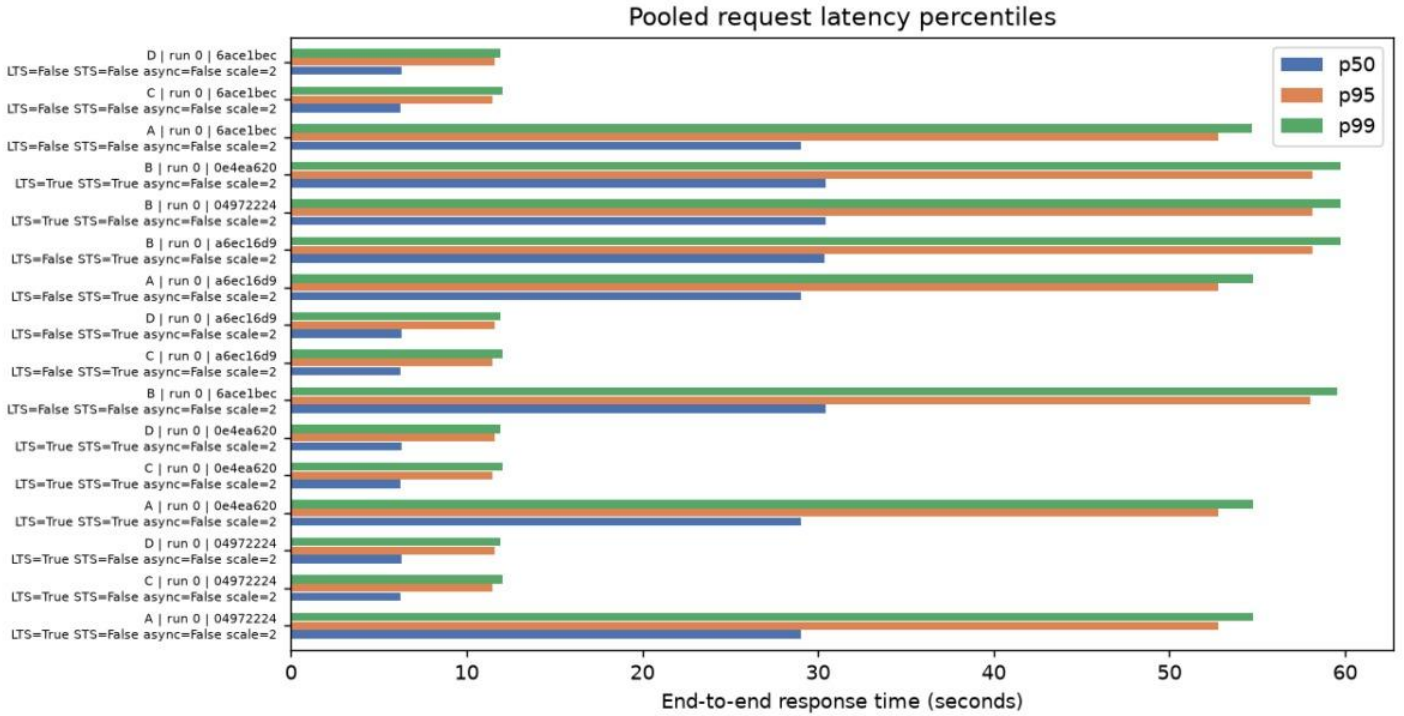
Major Plots in details



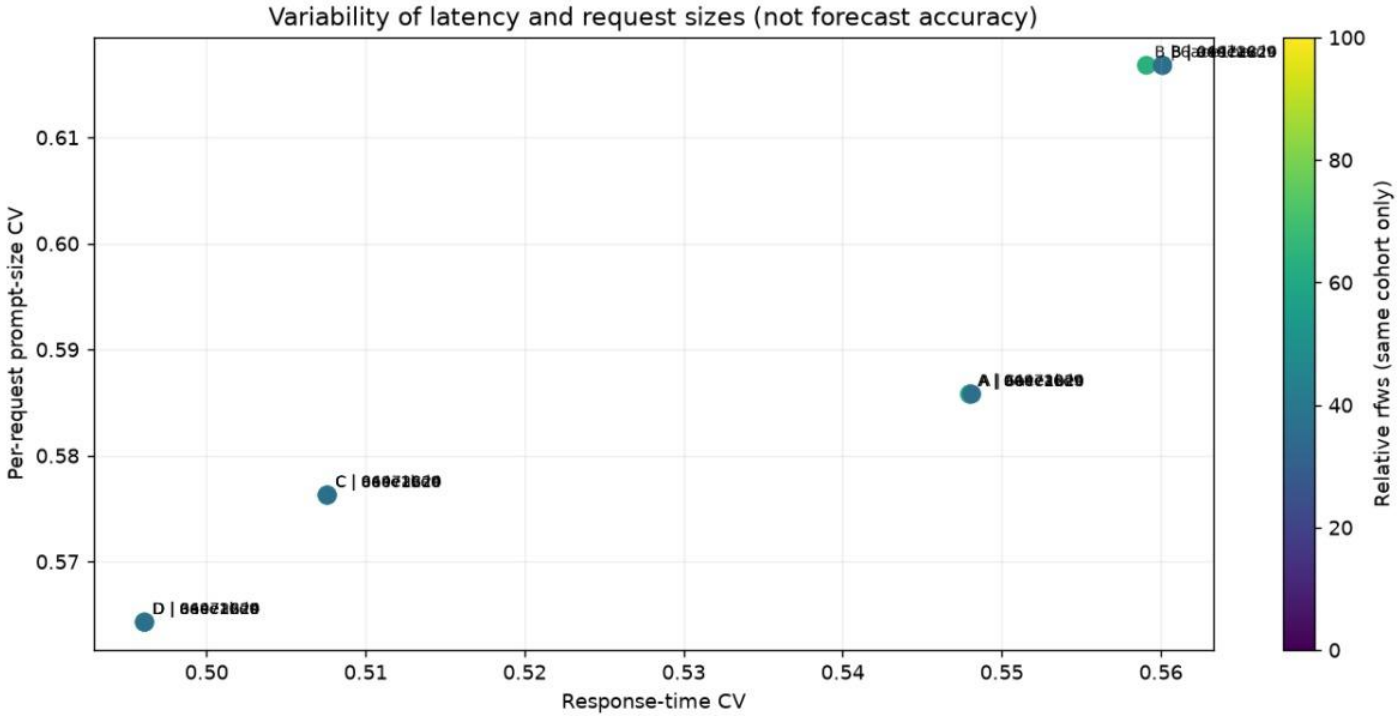
Major Plots in details



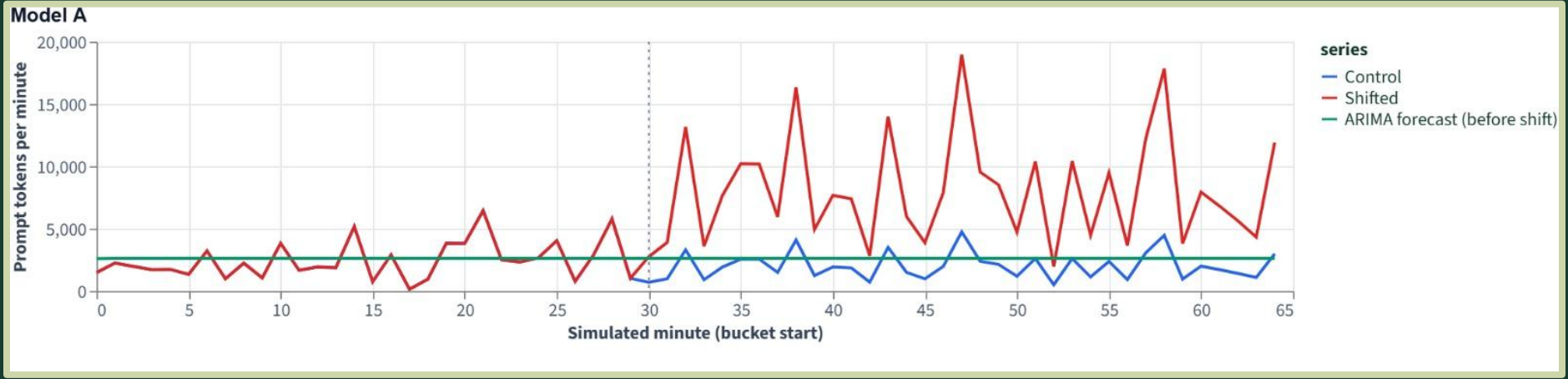
Major Plots in details

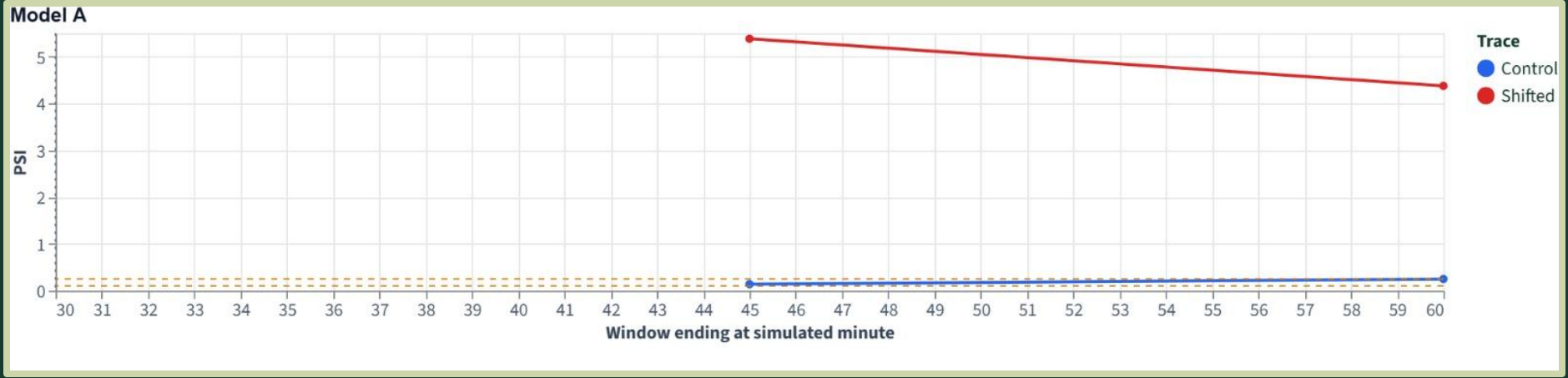


Major Plots in details



Drift Plots


Model A
Prompt tokens per minute
20,000
15,000
10,000
5,000
0
0
5
10
15
20
25
30
35
40
45
50
55
60
65
Simulated minute (bucket start)
series
Control
Shifted
ARIMA forecast (before shift)


Model A
PSI
5
4
3
2
1
0
30
31
32
33
34
35
36
37
38
39
40
41
42
43
44
45
46
47
48
49
50
51
52
53
54
55
56
57
58
59
60
Window ending at simulated minute
Trace
Control
Shifted

Module 8: Docker and environments

Purpose

Provides a reproducible runtime even though the tools need conflicting Python versions.

What it provides

The same setup on any machine, with the console on port 8501.

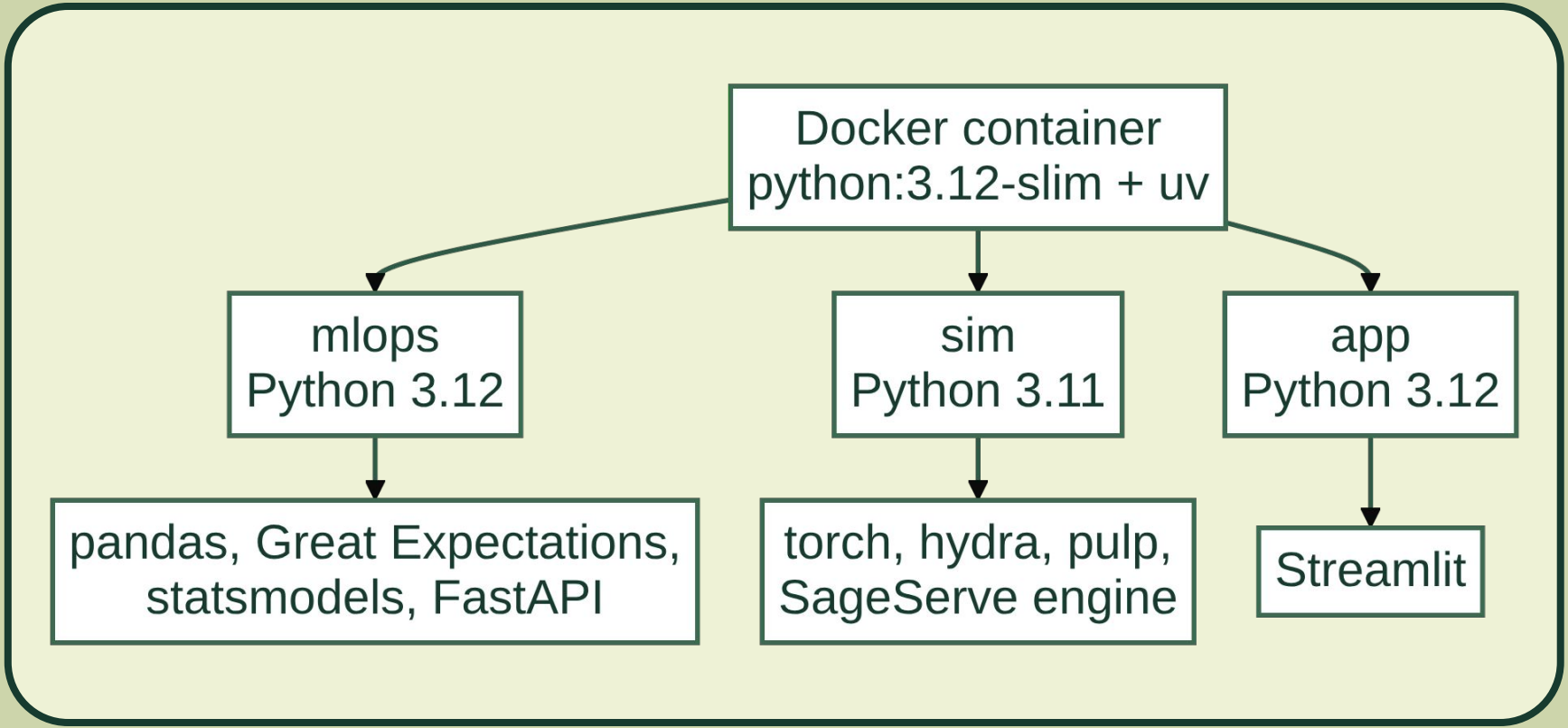
What it does

Builds three isolated environments in one image with uv, routes each command to the right one and runs a developer shell and the web console.

Why it matters

Dependencies never contaminate each other.

Container layout



Docker helps us create developer side CLI for us to run instructions, without worrying about setting up venvs and constant venv switching.

Docker also helps us to setup the host server as it mounts to the repo, which allows persistent disk writes.

```
docker compose up -d webui
```

```
[+] Creating 1/1
 ✔ Network sage-serve_default  Created
SageServe dev container
  mlops | sim | root          switch directory + venv
  registry <args>             model registry CLI      (e.g. registry list)
  pipeline <args>             pipeline runner         (e.g. pipeline --source static)
  latest <source> <stage>     latest artifact path    (e.g. latest streaming forecasts)
  run-sim <source>            simulator on latest trace + forecasts
  inference                   inference entry point
  sageserve-help              show this again
root@589a5b9a9983:/workspace#
```

Module 9: Streamlit Web App

Purpose

Gives engineers and operators one place to control and inspect the platform.

Why it matters

The whole workflow is available without the command line.

What it does

Lets them generate or stream data, run ETL, view lineage, inspect and promote models, launch simulations and explore results.

NOTE

The main objective of the project is the mlops pipeline not about the web interface.

Hence **Claude** was used to make the webapp module.

Module 10: Repository and CI/CD pipeline

Purpose

Git is used as a version control.

GitHub is a cloud-based hosting platform built on top of Git. Beyond storing your repository remotely, it provides tools for team collaboration, code reviews

Github Actions verifies that every change keeps the whole chain working.

What it does

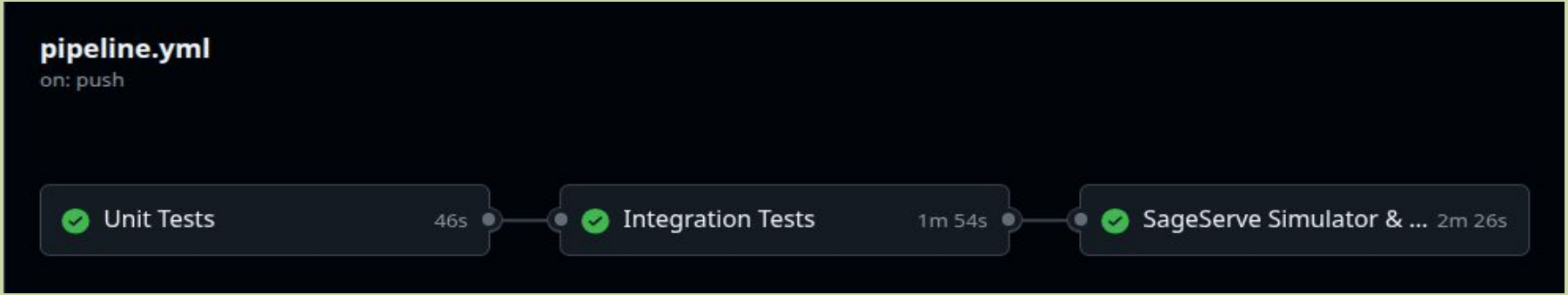
Trigger: Every **push** and **pull** request

Unit Tests: Fast isolation checks for core logic

Integration Tests: End-to-end lineage, pipeline, register & promote workflows

Simulator & Smoke Tests: Full operational & inference validation

Outcome: Zero breakages merged to main



Sage-Serve

Public

Pin | Watch 0 | Fork 0 | Starred 1

main | 2 Branches | 0 Tags | Go to file T | Add file | Code

SreenathKarthick11 Merge pull request #22 from SreenathKarthick11/sreen... 7918879 · 2 days ago | 137 Commits

.github/workflows	CI/CD : updated the pipeline.	2 days ago
SageServe-original	Docker : deployed the web app in my laptop when the co...	2 days ago
docker	Docker : Added docker env for code setup and easy frien...	5 days ago
docs	Docs : Updated current_run.md	5 days ago
mlops	Docker : Updated docker and core_code, so runs the web ...	3 days ago
notebooks	Added trace generator comparison notebook	5 days ago
scripts/plotting	Refactor : repository is refactored.	last month
.dockerignore	Docker : Added docker env for code setup and easy frien...	5 days ago
.gitignore	.gitignore : Updated.	5 days ago
Dockerfile	Docker : deployed the web app in my laptop when the co...	2 days ago
LICENSE	Initial commit	2 months ago
README.md	Initial commit	2 months ago
bundle.py	feat: bundle.py - bundles the code into a single plain file	last month
docker-compose.yml	Docker : Updated docker and core_code, so runs the web ...	3 days ago

README | MIT license

About

The Project for the Course : DS5619 MLOPS

gpu-monitoring | llm | llmops | mlops

Readme
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Activity
1 star
0 watching
0 forks

Releases

No releases published
Create a new release

Packages

No packages published
Publish your first package

Contributors 4

SreenathKarthick11
IJustKen
Simeonk1307 Simeon K.

Issues

New issue

is:issue

10 results | Author | Labels | Projects | Milestones | Assignees | Newest

- Integrating SageServe and MLOPs `mlops` `sage-serve`
  SreenathKarthick11/Sage-Serve#19 · SreenathKarthick11 opened 5d ago
- BUG : In the implementation of Model Factory `bug` `mlops` `model-regisitry` — 1
  SreenathKarthick11/Sage-Serve#17 · by SreenathKarthick11 was closed 5d ago
- Trace Generation `data` `mlops` `trace` — 1
  SreenathKarthick11/Sage-Serve#12 · by SreenathKarthick11 was closed 1mo ago
- ML for ILP / GPU Allocation `ml algorithm` `model` `question` — 1 — 1
  SreenathKarthick11/Sage-Serve#9 · by SreenathKarthick11 was closed 5d ago
- Implement Workload Drift and Forecasting Performance Monitoring `enhancement` `if-time-permits` `monitoring`
  SreenathKarthick11/Sage-Serve#8 · SreenathKarthick11 opened on Oct 1
- Implement CI Pipeline for Data, Forecasting and Simulator Validation `ci-cd` — 1 — 1
  SreenathKarthick11/Sage-Serve#7 · by SreenathKarthick11 was closed 4d ago
- Containerize the End-to-End SageServe MLOps Workflow `docker` `enhancement` — 1 — 2
  SreenathKarthick11/Sage-Serve#6 · by SreenathKarthick11 was closed 2d ago
- Set Up Forecasting Model Registry and Governance `enhancement` `ml algorithm` `model` `registory` — 1 — 1
  SreenathKarthick11/Sage-Serve#5 · by SreenathKarthick11 was closed 5d ago
- Implement Dataset, Forecast and Experiment Versioning with Lineage `data` `versioning` — 2
  SreenathKarthick11/Sage-Serve#4 · by SreenathKarthick11 was closed 1w ago
- Implement ETL Pipeline and Workload Trace Validation `data-validation` `enhancement` — 1
  SreenathKarthick11/Sage-Serve#3 · by SreenathKarthick11 was closed on Oct 7

All pull requests

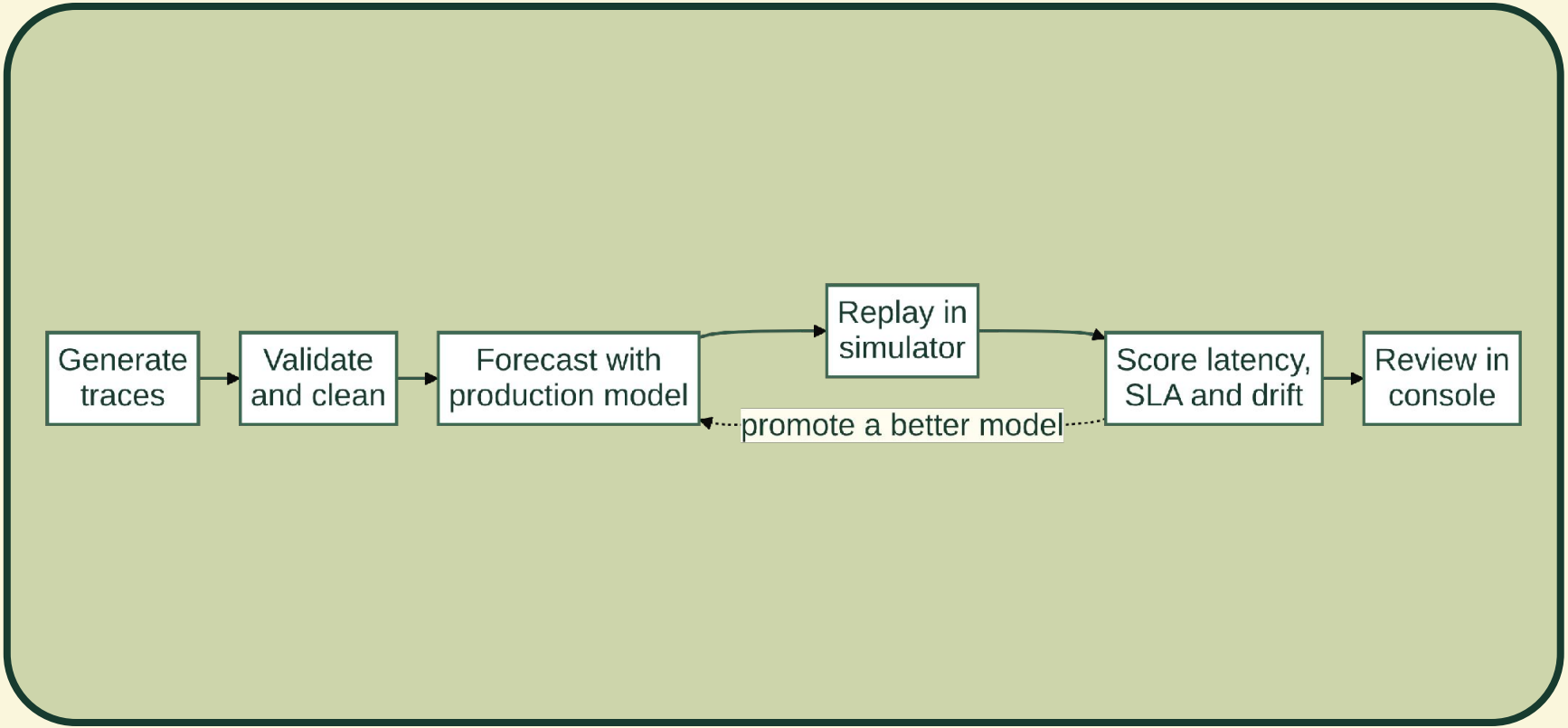
New pull request

is:pr

Open 1 Closed 12 Author Label Projects Milestones Reviews Assignee Newest

- Modify inference reporting and add allocation and drift monitoring
  #23 · AakashKiran opened 18 hours ago Collaborator · ✓ 3/3
- [Docker] : Containerize the Entire Application and Host it in a server. deployment docker sage-serve web-app
  #22 · by SreenathKarthick11 was merged 2 days ago Owner · ✓ 3/3
- [Web UI] : Added a Web App using Streamlit web-app
  #21 · by SreenathKarthick11 was merged 3 days ago Owner · ✓ 3/3
- CI/CD : Added Test and CI CD pipeline ci-cd test
  #20 · by SreenathKarthick11 was merged 4 days ago Owner · ✓ 6/6
- Updated inference validation and metrics, and added trace comparison notebook inference mlops
  #18 · by AakashKiran was merged 5 days ago Collaborator · Approved
- [Model Registry] : Added a Model registry mlops model model-registry
  #16 · by SreenathKarthick11 was merged 5 days ago Owner
- Simeon/versioning lineage
  #15 · by Simeonk1307 was merged last week Collaborator

End-to-end lifecycle



Thank you

Questions?